

Name :- Himanshu Heda

Class :- B MCA Roll No. :- 24225013

Sub :- NNDL

Copy - 1

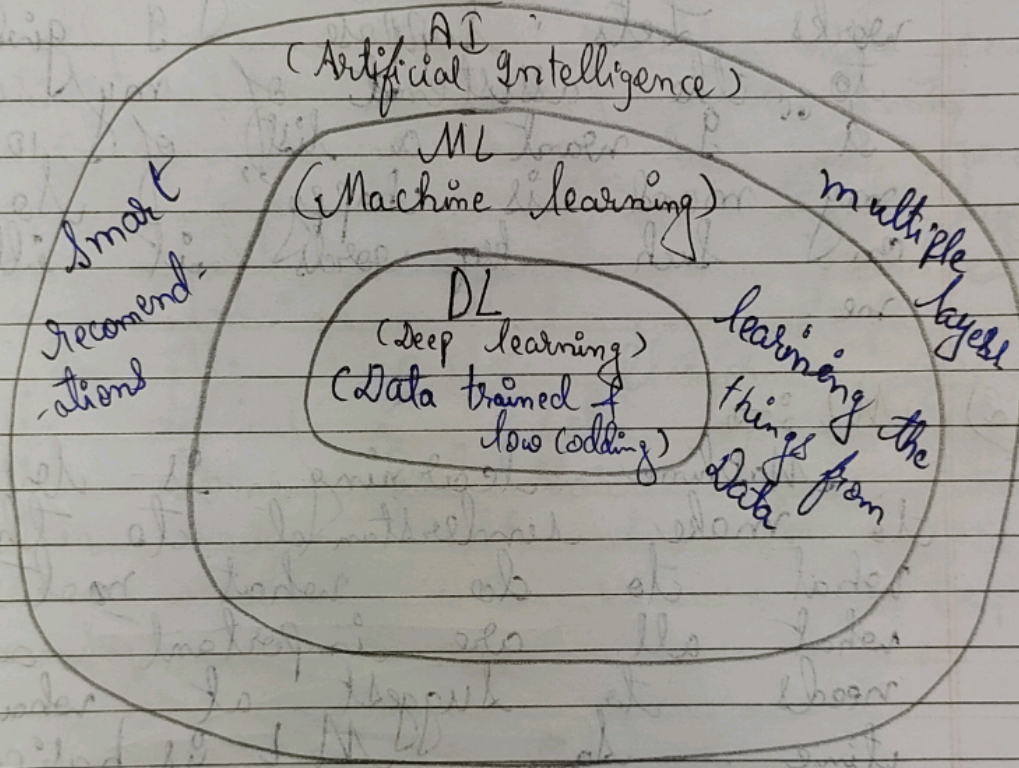
Page No.

Date

CIA-I

Q.1)
Sol.

A Company which build an OTT platforms like Netflix, Youtube, Amazon Prime, etc platforms will use the AI, ML & DL to the system to enhance its works are like the below :-



So let's Go deeper into it :-

1) AI :-

Artificial Intelligence is the most usable thing in today's world we all know the Chat GPT, Gemini, Grok type of platforms there are few AI models so.

for the OTT platforms like Netflix we can use some chat bot assistant to the user like if a user will ask for some particular movie recommendation or for the movies which are based on its current mood like the Instagram algorithm works. Let's suppose I give a prompt to the assistant of my OTT platform "I want a list of 10 movies my mood is happy" so based on such key words it will suggest me.

2) ML :-

Machine learning is typically means to make understood to the machine what to do what not to do what all are important and what needs to suggest at what time. So ML is basically the learning from a dataset provided to the models to train over so in the case of OTT platform I have given some proper prompt which it will convert to the keywords and based on the keywords like happy, mood, list of 10 movies it will filter out the movie list which are present

on our OTT platform and then show the recommendations on it.

3) DL :-

Deep learning it is basically the subset of AI & ML it will come after the AI & ML in this basically we will provide the smart learning by providing the huge data set it is having one advantage of multiple layers and will made to handle that layers part in deeper. In the OTT platform we basically defines a lot of things with some particular recommendation we needed so for that recommendation we use the Deep learning.

Q.2)

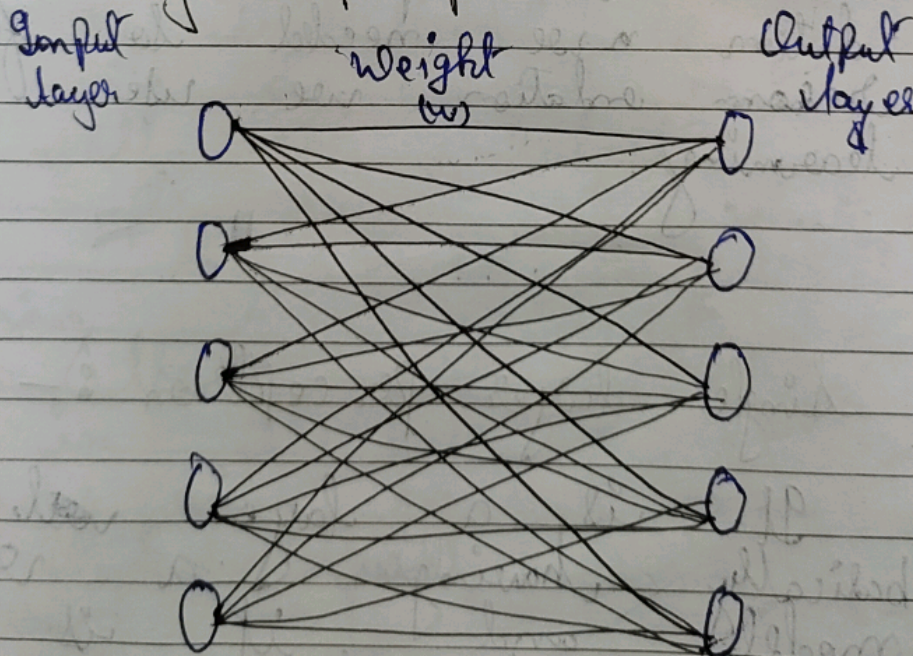
Solo

Single layer perceptron :-

It is a layer which is basically having a very easy model and it is having a best fit for the small or some short model taking apart from it it is not

a good recommendation to take
 and we are using $\oplus$
 XOR Gate which is having
 the two inputs and the
 output will be like the
 if 2 same input are there will
 result to have the output
 as 1 and if the inputs
 are different then the output
 would be 0 based on the XOR
 and single-layer we are doing
 this will surely leads to
 having a crash to the
 system.

Single layer perceptron :-



Single-layer

Himanshu Hecla
Roll No. :- 24225013

Copy - 2

Page No.

Date

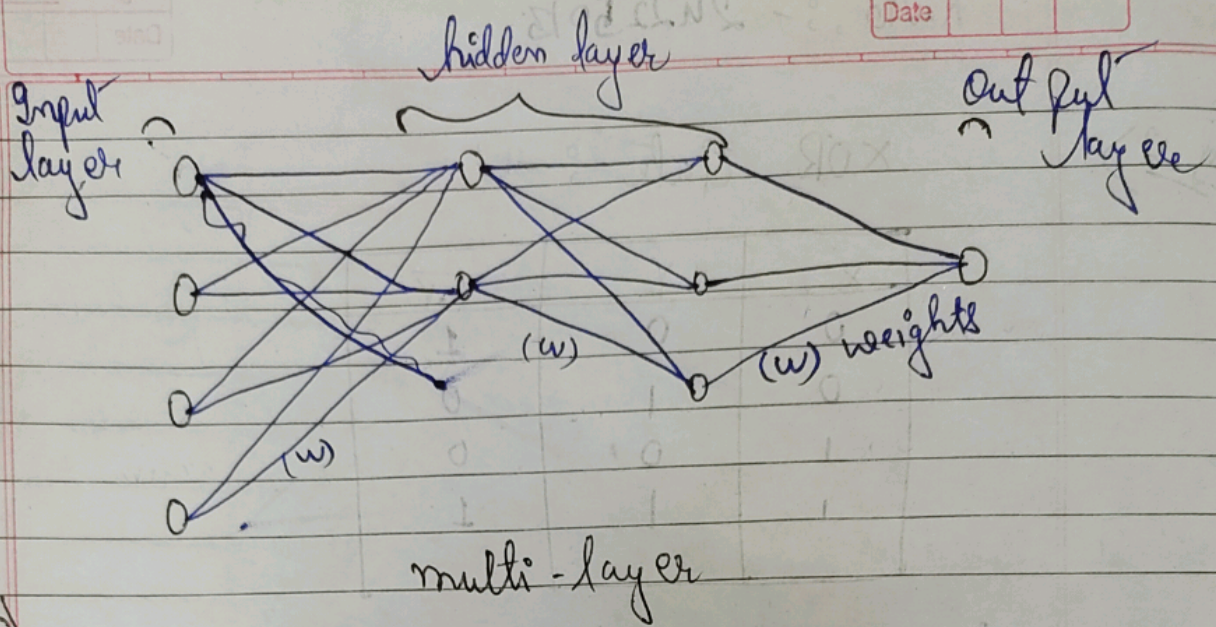
Q.2)

XOR Gate :-

x	y	$\overline{x \cdot y}$
0	0	1
0	1	0
1	0	0
1	1	1

So I would like to change the approach from single-layer perceptron to the multi-layer perceptron because it is having the advantage of multiple layers like the hidden layer. It will help to have the good calculation and have a complex approach as well. And there is one more layer ~~set~~ which is also having the best fit for this which is having the feed back options as well and a back propagation too to get the better and the best results.

$$\text{layers} = x_i \cdot w_i + b$$



Q.3)

Q.3)

ANN Model (Artificial Neural Network) :-

The model is a model which is use so ~~for~~ ~~train~~ train on the large dataset and to get the good results using the artificial intelligence. So if the accuracy is good and the results are not coming properly will ~~have~~ having various causes are like :-

1) heavy loss :-

If we having a good accuracy and having a ~~do~~ good amount of loss as well will leads to have the good accuracy but not the good results.

2) Loss is in the Increasing Stage:-
If we are continuously after every epoch the loss is increasing instead of decreasing also leads to have ~~and~~ not good results as per our recommendation.

3) Data biased :-
If the Dataset which we take is biased like not having a proper distribution of train, test & validation will lead to not good results.

4) Learning rate is too high :-
If the ~~learning~~ learning rate is getting too high like $\times 10^{-1}$ or more then ~~also~~ it will of some times lose lot of data and will not train properly.

5) Layer distribution :-
If we have not distributed the layer properly and will train our data will also provide some of the issues like this at the time of results.

Q.4)

1) Epoch :- $\rightarrow$

It is basically a distribution of the epoch in equal parts like having the input will lead the data on with the help of weight and then to the hidden layer and after that will go to the output will track the data with same process opposite and will provide the result will be completing a single epoch.

2) Learning Rate :-

It is a rate of learning the data with the help of given input in every epoch the proper distributed data will be learning which some rates which will provide higher the learning rate leads to have a good recommendation - i.e. the number which we will give & provide.

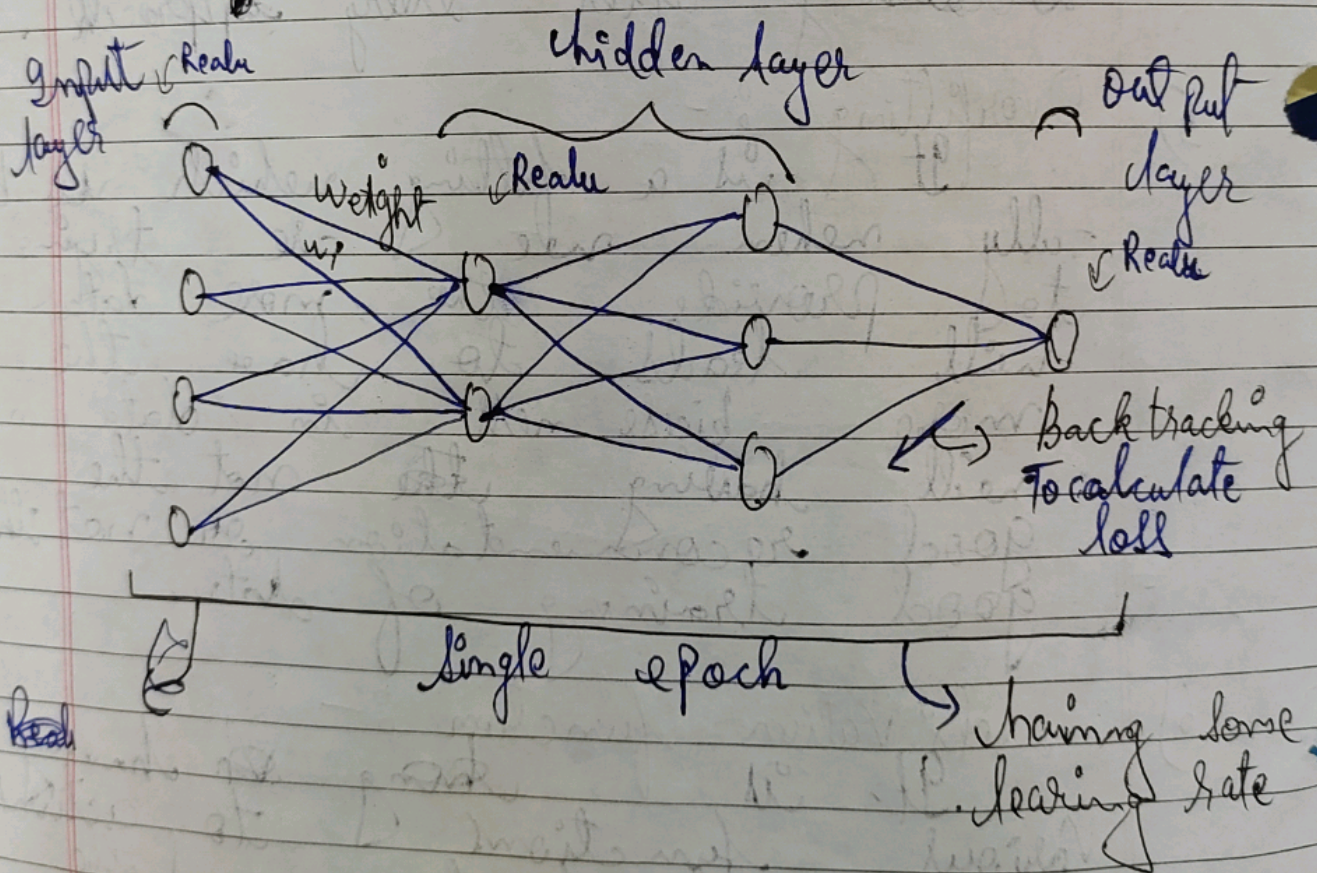
3) Loss function :-
It is basically the function which work during the backtracking and if the loss will be decreasing or less loss it there will lead to have a good model train and a good recommendation if the loss is less of decreasing after every approach.

5) Overfitting :-
It is a fitting which is basically when we are trying to provide the more data will lead to have the more biasness in data and will having the not the good recommendation or not the good training of data.

6) Activation function :-
It is having various functions to activate the input function and the hidden layer will use it as well and the output layer will be having good to use.
The most popular activation

A hand-drawn graph on lined paper. The vertical axis has tick marks at -1, 0, and 1. The horizontal axis has tick marks at 0 and 1. Two sigmoidal curves are plotted. The blue curve starts at (0, -1) and increases towards 1. The black curve starts at (0, 0) and increases towards 1, crossing the blue curve.

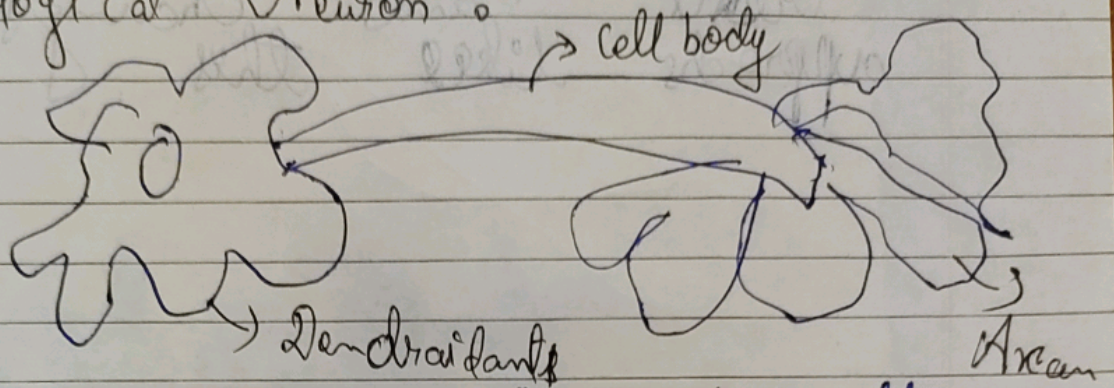
- sigmoid



Q.5)

Sol.

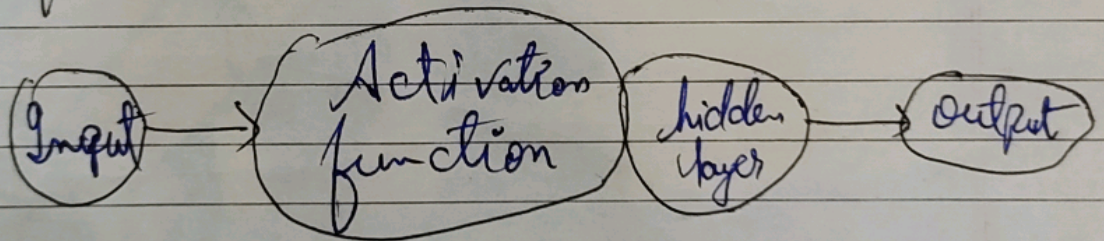
Biological Neuron :-



Dendrites → will
cell body → will
Axon → will

start the neuron
transfer the data
lead to have
recommendations.

Artificial Neuron :-



So the mixing of both the model will help to have a good robotic mimic to get the good results by training the data and so make learn and recommended results to get from the robotic and the expected results.

Today robots are also using this mixed approach to get the robots and changing the approaches like this.

